

 UNIVERSITY OF TECHNOLOGY - VNUHCM FACULTY OF COMPUTER SCIENCE AND ENGINEERING	HOMEWORK 5		Semester/Academic year		1	2026-2027
			Due		14/10/2026	
	Course title	Machine Learning				
	Course ID	CO3117				
	Duration	Take-home	Homework set		HW5	
Notes: Handwritten homework, submitted as homework/hw5-submission.pdf with tag hw5.						

Instructions

- **Genetic algorithms** (Chapter 5). Five problems, each worth 2 points (0.5 per part); total 10 points.
- **Write by hand**, on paper or with a stylus. Show your working: a bare number earns no credit. Put your full name, student ID and the set number at the top of the first page.
- **Work on your own.** Do not use AI tools, solution manuals or classmates' work before your submission is tagged. You may use your lecture notes and the course slides. Any later AI use goes in AI_USE.md, as for the weekly drills.
- **Submit in your course repository.** Scan all pages, in order, into one PDF of at most 5 MB, named homework/hw5-submission.pdf. Commit it, run python tools/check.py --checkpoint hw5, then git tag hw5 and git push origin main hw5.
- **Deadline: 14/10/2026 23:59** (the time GitHub receives the tag hw5). Solutions are released on 15/10/2026. Afterwards you may add homework/hw5-corrections.md; never replace or re-tag the submission.

Problem 1 (L.O.2) (2 pts): Answer the following questions about hypothesis representation and fitness in genetic algorithms. A rule is encoded as a bit string with one bit per attribute value, in this order: Outlook (Sunny, Overcast, Rain), Wind (Strong, Weak), and PlayTennis (Yes, No) for the conclusion.

- Encode the rule: IF Outlook = Sunny or Rain AND Wind = Strong THEN PlayTennis = Yes.
- What does the substring 111 mean for Outlook? What would the substring 000 mean, and why is it undesirable?
- Describe the mutation operator for bit strings and explain its role in the search.
- The course uses the fitness function $Fitness(h) = (correct(h))^2$, where $correct(h)$ is the fraction of training examples classified correctly. Compute the fitness of rules that classify 80% and 60% of the examples correctly, and explain the effect of squaring.

Problem 2 (L.O.2) (2 pts): Answer the following questions about mutation in genetic algorithms. A population has 50 chromosomes of 20 bits, and each bit is flipped independently with probability 0.01.

- What is the role of mutation in a GA?
- Compute the expected number of flipped bits per chromosome and per generation, and the probability that a given chromosome is not changed by mutation.
- What happens if the mutation probability is much too high, or much too low?
- Elitism copies the best one or few chromosomes of each generation unchanged into the next. Why can it be useful, and what is its risk?

Problem 3 (L.O.2) (2 pts): Answer the following questions about a genetic algorithm for the 0/1 knapsack problem. There are 8 items with weights (3, 4, 2, 5, 1, 6, 2, 3) and values (4, 5, 3, 8, 2, 7, 3, 4); the capacity is 12.

- Propose a chromosome representation. Decode the chromosome 10110010: which items are chosen, and what are the total weight and value?
- Propose a fitness function that handles overweight solutions with a penalty. Evaluate it for 11010100 using a penalty of 2 per unit of excess weight.
- Apply bit-flip mutation to the last bit of 10110010. Is the result feasible? What does this tell you about mutation and constraints?

D. In tournament selection, two hypotheses are drawn at random and the fitter one is selected with probability $p=0.75$ (otherwise the other one). If the pair drawn has fitness values -3 and 5 , what is the probability that each is selected? Why does this scheme still work when some fitness values are negative, as can happen with a penalty?

Problem 4 (L.O.2) (2 pts): Answer the following questions about encodings in genetic algorithms:

- A. Which encoding suits feature selection, and which suits the travelling salesman problem (TSP)? Explain.
- B. Why is bit-flip mutation unsuitable for a TSP tour written as a permutation? Apply swap mutation to the tour $(3\ 1\ 4\ 5\ 2)$, exchanging positions 2 and 5.
- C. A real variable x is encoded by 8 bits as $x=0.02k$, where k is the integer value of the bit string. Decode 10110010. What are the range and the resolution of this encoding?
- D. Order crossover (OX) for permutations keeps the segment of parent 1 between two cut points and fills the remaining positions, starting after the second cut point and wrapping around, with the missing cities in the order they appear in parent 2 from that point. Apply OX to $P_1 = (1\ 2\ 3\ 4\ 5\ 6\ 7\ 8)$ and $P_2 = (3\ 7\ 5\ 1\ 6\ 8\ 2\ 4)$ with cut points after positions 3 and 6.

Problem 5 (L.O.2) (2 pts): Answer the following questions about the use of genetic algorithms:

- A. Give two advantages of genetic algorithms mentioned in the course.
- B. The basic GA loop runs until the best fitness in the population reaches a given fitness threshold. Name two other practical termination criteria.
- C. Where does the inductive bias of a genetic algorithm come from?
- D. Design a GA for selecting features from a data set with 10 features: give the chromosome encoding and a fitness function that rewards accuracy but penalises using many features. Evaluate your fitness for the chromosome 0110010001 if the selected features give a validation accuracy of 0.86.

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